

Introduction to Quantum Computing - Week 1 Lesson Plan

1 Lesson Overview

Duration: 50 minutes

Goal: Introduce fundamental concepts of quantum computing, including qubits, bra-ket notation, tensor products, orthogonality, and basic quantum gates.

2 1. Introduction to Quantum Computing (10 min)

- **What is Quantum Computing?**
 - Classical vs. Quantum: Key differences.
 - Why study quantum computing? (Brief mention of real-world applications.)
 - The fundamental unit: the **qubit**.

- **What is a Qubit?**
 - Classical bits: 0 or 1.
 - Quantum bits: Linear combinations.
 - **Mathematical representation:**

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad a, b \in \mathbb{C}, \quad |a|^2 + |b|^2 = 1$$

- **Standard basis states:**

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

3 2. Mathematical Foundations (20 min)

3.1 Bra-Ket Notation (10 min)

- **Dirac notation and vector representation.**
- **Inner product:** $\langle 0|0\rangle = 1, \langle 0|1\rangle = 0$.
- **Outer product:** $|\psi\rangle\langle\phi|$ as a matrix.
- **Orthogonality:** $\langle 0|1\rangle = 0$, essential for measurement.

3.2 Matrix Multiplication Refresher (5 min)

- Applying a gate (matrix) to a qubit (vector).
- Example calculation:

$$I|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

3.3 Tensor Product (5 min)

- Needed for multi-qubit systems.
- Example:

$$|00\rangle = |0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- Demonstrating four-dimensional state space.

4 3. Basic Quantum Gates (15 min)

4.1 Bit Flip Gate: The Pauli-X Gate (10 min)

- Definition:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Effect on basis states:
 - $X|0\rangle = |1\rangle$, $X|1\rangle = |0\rangle$.
 - Similar to a classical NOT gate.
- Explicit matrix multiplication example.

4.2 The Identity Gate (5 min)

- Definition:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Leaves states unchanged, useful for multi-qubit operations.

5 4. Hands-on Examples Discussion (5 min)

- Compute $\langle 0|1\rangle$ and $\langle 1|1\rangle$.
- Compute $X|0\rangle$ and $X|1\rangle$.
- Compute $|0\rangle \otimes |1\rangle$.
- Open discussion for questions.

6 5. Wrap-up (Final 2 min)

- Recap: qubits, bra-ket notation, orthogonality, tensor products, bit flip gates.
- What's next: Quantum gates beyond bit flips.